Euphytica (2008) 159:191–198
DOI 10.1007/s10681-007-9472-9

Molecular mapping of genomic regions harboring QTLs for stalk rot resistance in sorghum

P. Srinivasa Reddy · B. Fakrudin · Rajkumar · S. M. Punnuri · S. S. Arun · M. S. Kuruvinashetti · I. K. Das · N. Seetharama

Received: 21 May 2006 / Accepted: 26 May 2007 / Published online: 23 June 2007
© Springer Science+Business Media B.V. 2007

Abstract Stalk rot, also called as charcoal rot in India, caused by *Macrophomina phaseolina*, is an economically important, soil borne disease in major sorghum growing areas across the world. A population of F_9 generation recombinant inbred lines (RILs), derived from IS22380 (susceptible) × E36-1 (resistant), along with parents were phenotyped in sick plots at two locations (Dharwad and Bijapur, Karnataka, India). A total of 85 polymorphic marker loci (62 nuclear and 4 genic SSRs, 19 RAPDs) was available for the construction of genetic map, spanning 650.3 cM in all the ten linkage groups. Analysis with QTL Cartographer (2.5b), adopting composite interval mapping method (LOD > 2.0) at both locations, revealed 5 QTLs at Dharwad and 4 QTLs at Bijapur locations for the component traits of charcoal rot disease resistance. QTLs for number of internodes crossed, length of infection and per cent lodging accounted for 31.83, 10.76 and 18.90 per cent at Dharwad location and 14.87, 10.47 and 26.44 per cent phenotypic variability at Bijapur location, respectively. The QTLs for number of internodes crossed by the rot, length of infection and percent lodging were common across two locations. These QTLs, consistent over environments for the component traits, are likely to assist in marker-assisted selection (MAS) for charcoal rot resistance in sorghum.

Keywords Stalk rot · Charcoal rot · *Macrophomina phaseolina* · Sorghum · QTL · Mapping

P. Srinivasa Reddy · B. Fakrudin · Rajkumar · S. M. Punnuri · S. S. Arun · M. S. Kuruvinashetti
Institute of Agri-Biotechnology, University of Agricultural Sciences, Dharwad 580005, India

Present Address:
B. Fakrudin (✉)
Division of Plant Sciences, Molecular Genetics and Soybean Biotechnology Laboratory, University of Missouri, Columbia, MO 65211, USA
e-mail: fakru_b@yahoo.com

I. K. Das · N. Seetharama
National Research Centre for Sorghum, Ranjendranagar, Hyderabad, India

Introduction

Sorghum [*Sorghum bicolor* (L.) Moench], mostly grown in marginal soils, is an important staple food for millions of poor people and valuable cattle feed in semi-arid tropics. At current global production of 58.9 m from 42.07 m ha, sorghum is next in importance to rice, wheat and maize as a food grain (FAO 2004). Because of its importance, an international team has sequenced the gene rich regions of sorghum recently and whole genome sequencing efforts are under progress (Bedell et al. 2005; Sorghum genomics planning workshop participants 2005; www.phytozome.net/sorghum). Considered as a coarse grain, it is rich in dietary fiber. Grain quality and productivity of sorghum, planted in post rainy season and grown under receding soil moisture conditions, is seriously affected by stalk rot (charcoal rot) disease caused by *Macrophomina phaseolina* (Tassi) Goid, a weak pathogen that is more active under water stress during grain maturity. The most striking indication of the disease is premature crop lodging, which is due to weakened stalk. The fungus normally enters the stem through the roots and disintegrates the pith, leaving unsupported vascular bundles covered with its black sclerotia (Maunder 1993). Retardation of plant vigour and metabolic changes due to drought predisposes the crop to attack by this, normally non-aggressive, pathogen (Edmunds 1964; Edmunds and Voight 1966; Odvody and Dunkle 1979). The extent of loss, mainly due to improperly filled grains, depends on the stage of infection. Dry fodder quality is invariably affected. Though different methods of screening for the disease are available, *in situ* (sick-plots) screening is considered to be the most realistic (Edmunds 1964). Due to complex quantitative inheritance of resistance to the disease, very little progress has been made in breeding for charcoal rot resistance (Mukury 1992; Rosenow 1992). Selection of stiff-stalk and non-senescent (stay-green) types with high productivity is considered important in breeding for charcoal rot resistance (Maunder 1993).

Modern biotechnology has provided DNA based molecular markers, heritable entities that might be genetically associated with economically important traits. In recent years, the availability of DNA markers and powerful biometric methods has lead to considerable progress in mapping of quantitative trait loci (QTLs) in crop plants. The most straight forward applications of QTL analysis are marker assisted selection (MAS) in breeding and pre-breeding and QTL cloning (Asians 2002). Availability of closely linked and easily scorable genetic markers to charcoal rot resistance can assist in breeding resistant cultivars and to understand the mechanism of resistance. Presently, MAS is considered as an important strategy for improving quantitative traits (Asians 2002). The components of resistance to charcoal rot in sorghum being quantitatively inherited, it is important to map related QTLs and mobilize them through MAS. Due to high genotype × environment (G × E) interaction, genetic analysis of charcoal rot resistance requires phenotyping in replicated trials in multiple environments to estimate the genotypic variance. Being a set of random and homozygous products of meiotic segregation, recombinant inbred lines (RILs) are very handy for phenotypic evaluation and genetic mapping with a variety of molecular markers over time and space.

Various genetic maps of sorghum have been constructed using RFLP (Peng et al. 1999; Xu et al. 2000), AFLP (Boivin et al. 1999), nuclear SSR (Taramino et al. 1997; Bhattramakki et al. 2000), RAPD (Tuinstra et al. 1996, 1997) and genic SSR (Punna et al. 2006) markers. A reasonably dense linkage map, in a specific purpose mapping population, is essential for any effective MAS program. Since, there are no published reports of QTLs for the charcoal rot disease in sorghum, an attempt was made to detect and map QTLs for the component traits of the disease.

Materials and methods

Plant material

The mapping population developed from a cross between a Sudanese dwarf line, susceptible for charcoal rot and a landrace of Ethiopian origin, resistant to charcoal rot, E36-1, consisted of 93 RILs (F_9 generation). The population was developed at ICRISAT, Hyderabad, Patancheru, India.

Molecular markers

In all 532 markers were considered for genotyping, which consisted of 262 nuclear SSR, 20 genic SSR and 250 Operon 10 mer primers (RAPD) markers. Selection of nuclear SSR markers was based on the linkage maps of Brown et al. (1996), Bhattramakki et al. (2000), Taramino et al. (1997) and Haussmann et al. (2002, 2004). The genic SSRs (with prefix iabt) were developed at the Institute of Agri-Biotechnology, India by mining publicly available sorghum ESTs at NCBI (Arun 2006). Genotyping of SSR markers was done with either 2% agarose (by ethidium bromide staining) or 6% polyacrylamide gels (PAGE) in a manual DNA sequencing apparatus (Bio–rad, USA) and visualized following silver staining. RAPD bands were separated on 1.5%

agarose gels and visualized by ethidium bromide staining under UV-transilluminator (UVi-Tech, England). Consistent polymorphic random decamer primers short listed by repeated screening were only employed for genotyping. In the map, the markers prefixed by one or two letters are RAPD markers. A total of 85 polymorphic markers were employed to generate marker segregation data across RILs.

Multilocation evaluation of RILs

RILs (93), their parents and checks-including a prominent cultivar, M35-1, were evaluated at two locations viz., main agricultural research station (MARS), Dharwad, and regional agricultural research station (RARS), Bijapur under natural, receding moisture conditions of the post-monsoon period (*rabi*). Simple lattice design was adopted for the trial, on an area of 25×21 m^2 with four replications. A replication consisted of 10 blocks, each with 10 lines, planted randomly. Each RIL was planted in a single row with 15 plants (row spacing 45 cm and between plants 15 cm) in the month of October 2004. In both locations, typical to *rabi* season of the region, there were no rains from pre-flowering stage of the crop till the end: a typical receding moisture situation which is congenial for the stalk rot manifestation. In order to eliminate the border effect, six rows of M35-1 were sown on all sides of the field.

Disease evaluation and scoring

Evaluation for charcoal rot incidence was done under natural sick-plot conditions. Observations were made 20 days after complete grain filling (complete maturity) on three parameters viz., per cent lodging on plot (15 plants per replication) basis and the number of internodes crossed by the rot and length of infection on individual plant basis. The number of lodged plants at the time of observation was counted. The stems of all plants were slit open to measure the length of infection (cm) and the number of internodes crossed by the rot.

Mapping of QTLs

Standard analysis of variance for each parameter was done for pooled and individual location phenotypic data, separately. Phenotypic (PCV) and genotypic coefficient of variation (GCV) and correlations were calculated according to Singh and Chaudhary (1977). The goodness of fit for segregation of marker loci in RILs was checked using chi-square test. MAPMAKER, version 3.0b (Lander et al. 1987) was employed for construction of the genetic linkage maps. Linkage between two marker loci was declared significant, in a two-point analysis, when the LOD score (log10 of the likelihood odds ratio) exceeded the threshold of 3.0. After identifying the linkage groups and the linear arrangement of loci, linkage distances were estimated by multi-point analysis by applying Haldane's mapping function (Haldane 1919). For mapping QTLs and estimation of their effects, the method of composite interval mapping (CIM) (Zeng 1993, 1994) was employed, using QTL Cartographer version 2.5b (Zeng 1993; Basten et al. 1994), separately for each location data.

Results

Phenotypic analysis

A continuous distribution of phenotypes for disease parameters at both Dharwad and Bijapur locations indicated quantitative trait. The data for all parameters followed normal distribution and it was used directly for analysis without transformation (Fig. 1). There were significant differences for charcoal rot parameters between parents and among RILs (Table 1). Transgressive segregation was also observed for all charcoal rot parameters. The number of internodes crossed by the rot had significant positive correlation with length of infection at both genotypic and phenotypic levels. Plant height was positively correlated with per cent lodging at both genotypic and phenotypic levels (Table 2).

Molecular linkage analysis

Parental lines were screened with 262 genomic SSR, 20 genic SSR and 250 RAPD markers. Between the parents, 62 nuclear SSR (23%), 4 genic SSR (20%) and 19 RAPD (7.9%) markers were polymorphic. All polymorphic markers were employed to screen RILs to generate genotypic data. After confirming the expected 1:1 ratio for alleles at each marker locus by chi-square test, the genotypic data was used to generate framework map for QTL analysis. Eight

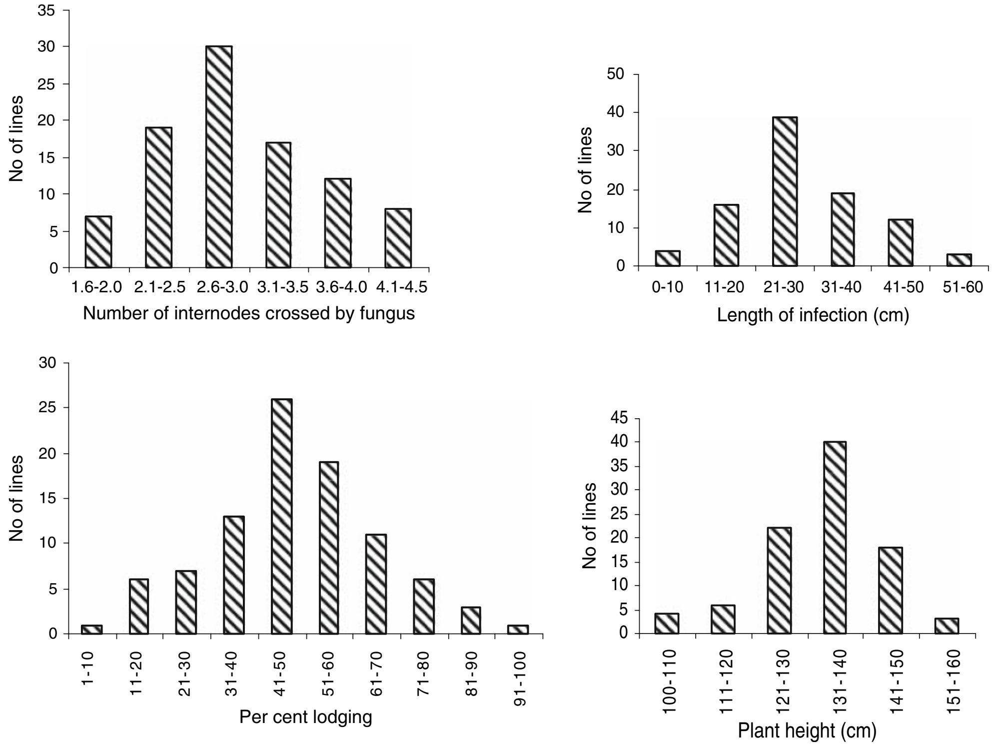

Fig. 1 Frequency distribution of RILs of IS22380 × E36-1 for three stalk rot resistance component traits viz., number of internodes crossed, length of infection and per cent lodging and plant height

Table 1 Components of variation for charcoal rot disease parameters in RILs of IS22380 × E36-1

Trait	Population parameters						
	Grand mean	Range	*G*	*G* × *E*	Error	S.Em.±	Heritability (h^2)
X1	2.95	1.75–4.25	1.287**	1.44**	0.147	0.284	55.72
X2	27.93	3.97–52.29	343.31**	258.35**	12.47	4.63	77.66
X3	48.49	10–100	2521.34**	41.68**	28.60	12.55	82.66
X4	132.74	100.63–152.16	796.00**	133.00*	99.10	4.97	52.47

*, **: Significant at 5% and at 1% probability level respectively

Note: X1: No. of internodes crossed by the rot, X2: Length of infection (cm), X3: Lodging per cent, X4: Plant height

Table 2 Phenotypic and genotypic correlation coefficients among charcoal rot disease component traits in RILs of IS22380 × E36-1

	X1	X2	X3	X4
X1	1	0.261**	0.12	−0.078
X2	0.288**	1	0.014	0.096
X3	0.168	0.027	1	0.363**
X4	0.168	0.027	0.365**	1

** Significant at 1% probability level

Note: X1: No. of internodes crossed by the rot; X2: Length of infection (cm); X3: Lodging per cent; X4: Plant height

Values above and below the diagonal are phenotypic and genotypic correlation coefficients, respectively markers were unlinked and the remaining 77 loci mapped within 10 linkage groups (Fig. 2), with 3–18 markers on each linkage group, spanning a length of 650.3 cM. The average distance between pairs of markers was 8.45 cM. The markers that showed segregation distortion were AB4, AT9, G17, H3, M17 (RAPDs) and xtxp290 xtxp354 (SSRs).

Mapping of QTLs for charcoal rot resistance

The method of composite interval mapping was applied to locate QTLs for charcoal rot using QTL Cartographer V 2.5b. At LOD 2.0, five QTLs at Dharwad and four QTLs at Bijapur locations were detected for the disease parameters (Table 3, Fig. 2). The QTLs for charcoal rot incidence were localized on A, B, D and I linkage groups of sorghum. Interestingly, two major QTLs on linkage group B and D were identified for the number of internodes crossed by the rot, which together accounted for 31.83 per cent of the total phenotypic variation at Dharwad location. At Bijapur, a single QTL on linkage group B was detected, which contributed to 14.87 per cent phenotypic variation. For the length of infection, only one QTL was detected on linkage group A, at both locations. Two QTLs were detected for per cent lodging on linkage group D and I, together accounting for 18.90 per cent of phenotypic variation at Dharwad location. At Bijapur, two QTLs on linkage group D were detected, which together contributed to 26.44 per cent of the phenotypic variation. Three major QTLs, one each for the number of internodes crossed by the rot, the length of infection and per cent lodging were common for both the locations.

Discussion

Stalk rot disease of sorghum is associated with premature stem lodging and pith disintegration,

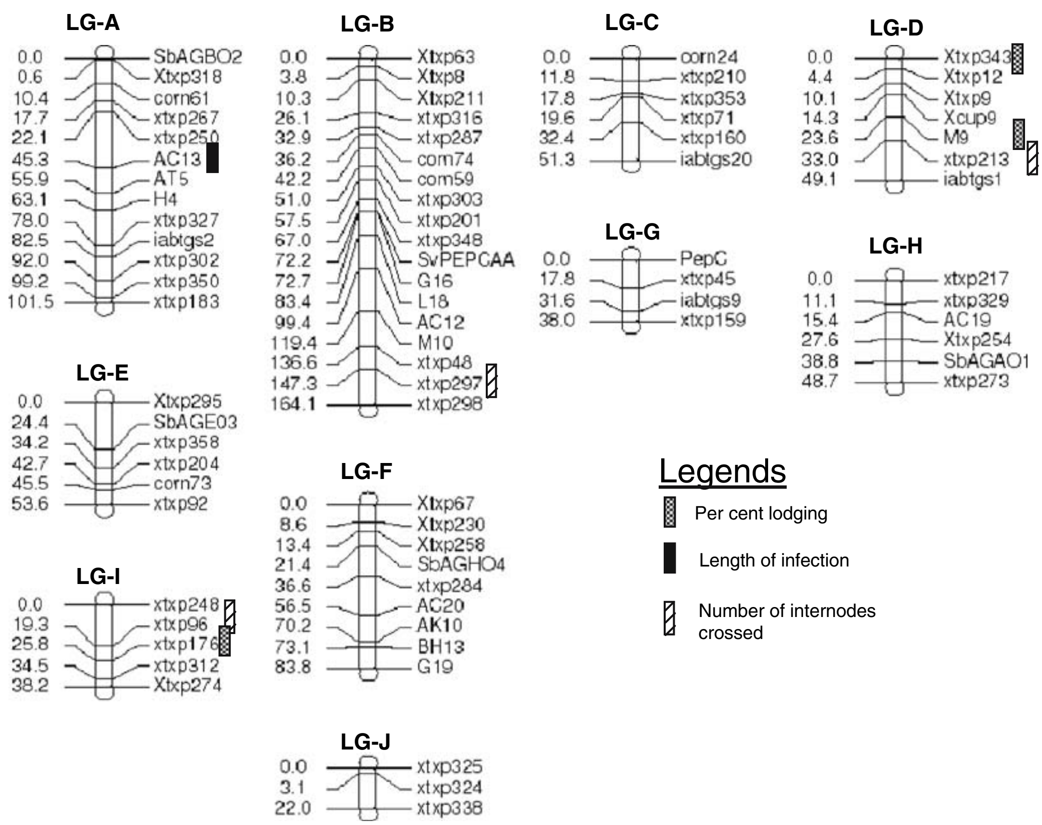

Fig. 2 Localization of QTLs for thee major component traits of charcoal rot-length of infection, per cent lodging and number of internodes crossed by fungus in a genetic linkage map developed for RIL population derived from IS22380 × 36-1 cross of Sorghum

Table 3 Features of QTLs for charcoal rot disease parameters in Dharwad and Bijapur locations

Traits	Marker	Linkage group	QTL position	Peak LOD²	Additive effect	Phenotypic variance
Dharwad						
No. of internodes crossed	**xtxp297**	**B**	163.3	4.4798	0.3934	19.29
	xtxp213	D	47.01	2.7918	0.2758	12.54
Length of infection	**AC13**	**A**	51.27	2.2607	2.1356	10.76
Percent lodging	**xtxp343**	**D**	0.01	3.0673	9.8628	11.01
	xtxp176	I	25.85	2.194	6.3081	7.89
Bijapur						
No. of internodes crossed	**xtxp297**	**B**	163.3	3.2124	0.3381	14.87
Length of infection	**AC13**	**A**	53.27	2.314	2.0842	10.47
Percent lodging	**xtxp343**	**D**	0.01	4.2728	12.6915	15.2
	M9	D	23.64	3.2533	7.9918	11.24

Common QTLs across locations are in bold letters

leading to inferior grain and fodder quality. Intensity of rotting varies widely depending on soil moisture and weather parameters at the time of grain filling and maturity. Inheritance of charcoal rot disease is believed to be complex and polygenic, often associated with drought tolerance (Tenkouano et al. 1993). The plant physiological status and prevailing soil moisture status determine the severity of the disease (Tesso et al 2005). The disease does not kill the plant and the losses due to this disease are primarily from loss in grain weight and poor quality fodder. Consistency in phenotyping for the disease is critical in breeding efforts and genetic investigation. Phenotypic assessment under artificial conditions is often not well correlated with data from field evaluation. In this study, material was evaluated in sick-plots maintained for many years, with yearly addition of inoculum in the form of infected stover.

There was significant variation for all the three disease parameters among the RILs, as expected, since the RILs were derived from deliberately selected, phenotypically distinct parents for a number of characters, including reaction to charcoal rot fungus. Genetic correlation showed that per cent lodging had significant positive correlation with plant height. Similar observations have been made in sorghum (Esechie et al. 1977) earlier. Number of internodes crossed by the rot had positive correlation with length of infection, which is expected, but these two parameters were not correlated with per cent lodging, an important parameter of disease manifestation and loss due to this factor. Though plant height is unlikely to be biologically related to the spread of infection through the stem and the number of internodes crossed by the rot, it was positively correlated with lodging, which happens to be one of the final manifestations of the disease.

Among 240 RAPD, 265 nuclear SSR and 20 genic SSRs screened, 85 (19 RAPD, 62 nuclear SSR and 4 genic SSR) markers that detected polymorphism between the parents were used for genotyping RILs. RAPD markers are considered less reliable and the genic SSRs primers are reported to be less polymorphic compared to nuclear SSRs in crop plants, because of greater DNA sequence conservation in transcribed regions (Scott et al. 2000; Russell et al. 2004). Finally, 77 marker loci were used to generate framework map for QTL analysis and eight were unlinked. Though the distances between the markers varied, the order of the common markers was the same as in the map of Bhattramakki et al. (2000), Haussmann et al. (2004) and spanned a total length of 650.3 cM. Majority of the markers mapped close to one another within a linkage group. Any QTL in these regions is likely to be closely linked to the markers. It was interesting to note that among the eight unlinked markers, six represented RAPD markers and two nuclear SSR markers. The random polymorphic markers eventually used for genotyping were selected through a stringent test for repeatability. Hence, the inherent nature of the very marker systems and size of the mapping population might influence such an observation.

Composite interval mapping (CIM) procedure employed here is one of the most recommended tools for QTL characterization as it uses multiple regression and background markers to avoid ghost QTLs. In the present investigation, for three parameters, 5 QTLs at Dharwad and 4 at Bijapur locations were detected on linkage groups A, B, D and I. These QTLs were associated with xtxp213, xtxp343 and M9 on D, AC13 on A, xtxp297 on B and xtxp176 on I. Linkage group D seems to be important as it has two major QTLs, one each for the number of internodes crossed by the rot and per cent lodging at Dharwad location and two QTLs for per cent lodging at Bijapur. Notably, three QTLs, one for each parameter, were common to both locations. Two of these QTLs associated with AC13 and xtxp343 on linkage group A and D were the same as those we detected previously with limited amount of marker data (Rajkumar 2004). As the flanking markers define the QTLs, the detection of QTL itself and its contribution to phenotypic variation is a function of the distance between the causative gene/s in the region and the markers that detected them. Thus, accurate detection of any QTL would require a high-density marker map. Until sorghum genetic marker maps become denser with the addition of more markers, this kind of discrepancies may be unavoidable. The additional QTLs detected in the present study compared to those reported by Rajkumar (2004), in the same populations, may be because of QTL analysis method applied and relatively larger number of markers used in present study. However, further saturation of the present linkage map with additional makers is highly desired and in progress. In any case, availability of consolidated physical and genetic reference map of sorghum would further facilitate the cross comparison between published sorghum genetic linkage maps (Haussmann et al. 2002).

Three major QTLs associated with markers xtxp297, AC13 and xtxp343, commonly detected between two locations, can be considered as stable. These, so-called ‘stable QTLs’ identified in this study may provide the basis for further evaluation and fine mapping of this region using genomic resources and their subsequent use in marker based breeding programs for charcoal rot resistance. However, further phenotyping at more number of representative locations is desirable to confirm the stability of these and other QTLs before using the linked markers for introgression breeding with MAS. Additionally, consideration of biochemical parameters such as sugars, phenols and other secondary metabolites seem to be relevant for the deep understanding of the stalk rot manifestation in sorghum. Developing additional mapping populations with diverse resistance and susceptible sources of charcoal root with substantial size would also help in dissecting the genetics nature of the trait and identification QTLs. Detailed characterization of these QTLs by identifying genes residing around them may also help in understanding the molecular basis of charcoal rot resistance in sorghum.

Acknowledgments This work was partly funded by Department of Biotechnology, Government of India and Ministry of Information Technology and Biotechnology (IT&BT), Government of Karnataka, India. We acknowledge the expert help of Dr. Y. D. Narayan and Dr. Palakshappa in disease scoring and fungal isolation. We thank Dr. Nandi Nagaraj, Dow Agro pvt. ltd., Bangalore for helpful suggestions on QTL analysis and Dr. B. D. Biradar and Dr. G. Sajjanar for their help during the evaluation at Bijapur location

References

Arun SS (2006) *In silico* EST data mining for elucidation of repeats biology and functional annotation in sorghum [*Sorghum bicolor* (L.) Moench.]. M. Sc. (Agri.) Thesis, University of Agricultural Sciences, Dharwad

Asians MJ (2002) Present and future of quantitative trait locus analysis in plant breeding. Plant Breed 121:281–291

Basten CJ, Weir BS, Zeng ZB (1994) Zmap-a QTL cartographer. In: Smith C, Gavora JS, Chesnais BBJ, Fairfull W, Gibson JP, Kennedy BW, Burnside EB (eds) Proceedings of the 5th world congress on genetics applied to livestock production: computing strategies and software, vol. 22. Guelph, Ontario, Canada, pp65–66. Organizing Committee, 5th World Congress on Genetics Applied to Livestock Production

Bedell JA, Budiman MA, Nunberg A, Citek RW, Robbins D, Jones J, FlicK E, Rohlfing T, Fries J, Bradford K, Mcmenamy J, Smith M, Holeman H, A Roe B, Wiley G, Korf IF, Rabinowicz PD, Lakey N, Mccombie WR, Jeddeloh JA, Martienssen RA (2005) Sorghum genome sequencing by methylation filtration. Public Library Sci Biol 3:e13

Bhattramakki D, Dong J, Chhabra AK, Hart GE (2000) An integrated SSR and RFLP linkage map of sorghum (*Sorghum bicolor* (L.) Moench). Genome 43:988–1002

Boivin K, Deu M, Rami JF, Trouche G, Hamon P (1999) Towards a saturated sorghum map using RFLP and AFLP markers. Theor Appl Genet 98:320–328

Brown SM, Hopkins MS, Mitchell SE, Senior ML, Wang TY, Duncan RR, Gonzalez-Candelas F, Kresorich S (1996) Multiple methods for the identification of polymorphic

simple sequence repeats (SSRs) in Sorghum (*Sorghum bicolor*). Theor Appl Genet 93:190–198
Edmunds LK, Voight RL (1966) Role of seed production in predisposition of sorghum to charcoal rot. Phytopathology 56:876
Edmunds LK (1964) Combined relation of plant maturity, temperature and soil moisture to charcoal stalk rot development in grain sorghum. Phytopathology 54:514–517
Esechie HA, Maranville JW, Ross WM (1977) Relationship of stalk morphology and chemical composition to lodging resistance in sorghum. Crop Sci 17:609–612
Food and Agriculture Organization (2004) Production estimates and crop assessment division. USDA, p 28
Haldane JBS (1919) The combination of linkage values, and the calculation of distance between loci of linked factor. J Genet 8:299–309
Haussmann BI, Hess DE Omanya GO, Folkertsma RT, Reddy BV, Kayentao M, Welz HG, Geiger HH (2004) Genomic regions influencing resistance to parasitic weed *Striga hermonthica* in two recombinant inbred populations of sorghum. Theor Appl Genet 109:1005–1016
Haussmann BI, Hess DE, Seetharama N, Welz HG, Geiger HH (2002) Construction of a combined sorghum linkage map from two recombinant inbred populations using AFLP, SSR, RFLP and RAPD markers and comparison with other sorghum maps. Theor Appl Genet 105:629–637
Lander ES, Green PL, Abrahamson J, Barlow A, Daly MJ, Lincoln S, Newburg L (1987) Mapmaker: an interactive computer package for constructing primary genetic linkage maps of experimental and natural populations. Genome 1:174–181
Maunder AB (1993) Breeding for stalk rot resistance as a component of acceptable agronomic performance. In: Sorghum root and stalk rots, Critical Review. Proceedings of the consultative group discussion of research needs and strategies for control of sorghum root and stalk rot diseases, 27 November–2 December 1983, Bellagio Italy. ICRISAT, Patancheru, Andhra Pradesh, 502324 India, pp219–224
Mukury SZ (1992) Breeding for grain mold resistance. In: de Milliano WAJ, Frederiksen RA, Bengaston GD (eds) Sorghum and millets disease: a second world review. ICRISAT, Patancheru, Andhra Pradesh 502 324, India, pp 273–285
Odvody GN, Dunkle LD (1979) Charcoal stalk rot of sorghum: effect of environment on host parasitic relations. Phytopathology 69:250–254
Peng Y, Schertz KF, Cartinhour S, Hart GE (1999) Comparative genome mapping of *Sorghum bicolor* (L.) Moench. using an RFLP map constructed in a population of recombinant inbred lines. Plant Breed 118:225–235
Punna R, Bantte K, Kumar ChA, Jayashree B, Folkertsma RT, Senthilve S, Mahalakshmi V, Reddy LA, Fakrudin B, Hash CT (2006) Development and mapping of EST-SSR markers in sorghum for comparative mapping with rice. In: Plant and animal genome (14–18, Jan 2006). San Diego, USA
Rajkumar (2004) Genetic and molecular analyses of charcoal rot resistance in sorghum. M. Sc. (Agri.) Thesis, University of Agricultural sciences, Dharwad
Rosenow DT (1992) Using germplasm from the world collection in breeding for disease resistance. In: de Milliano WAJ, Frederiksen RA, Bengston GD (eds) Sorghum and Millers Diseases—A second world review. ICRISAT, Patancheru, Andhra Pradesh, 502324, India, pp 319–324
Russell J, Booth A, Fuller J, Harrower B, Hardly P, Machray G, Powell W (2004) A comparison of sequence based polymorphism and haplotype content in transcribed and anonymous regions of the barley genome. Genome 47:389–398
Scott KD, Eggler P, Seaton G, Rossetto M (2000) Analysis of SSRs derived from grape ESTs. Theor Appl Genet 100:723–726
Singh RK, Chaudhary BD (1977) Biometrical methods in quantitative genetic analysis. Kalyani Publishers, New Delhi, India
Sorghum Genomics Planning Workshop Participants (2005) Towards sequencing of the sorghum genome. A U.S. national science foundation-sponsored workshop report. Plant Physiol 138:1898–1902 and also see http://www.phytozome.net/sorghum
Taramino G, Tarchini R, Ferrario S, Lee M, Pe ME (1997) Characterization and mapping of simple sequence repeats (SSRs) in *Sorghum bicolor*. Theor Appl Genet 95:66–72
Tenkouano A, Miller FR, Frederiksen RA, Rosenow DT (1993) Genetics of nonsenescence and charcoal rot resistance in sorghum. Theor Appl Genet 85(5):644–648
Tesso TT, Claflin LE, Tuinstra MR (2005) Analyses of stalk rot resistance and genetic diversity among drought tolerant sorghum genotypes. Crop Sci 45:645–652
Tuinstra MR, Ejeta G, Goldsbrough PB (1997) Heterogenous in bred family (HIF) analysis: an approach for developing near—isogenic lines that differ at quantitative trait loci. Theor Appl Genet 95:1005–1011
Tuinstra MR, Grote EM, Goldsbrough PB, Ejeta G (1996) Identification of quantitative trait loci associated with preflowering drought tolerance in sorghum. Crop Sci 36:1337–1344
Xu W, Subudhi PK, Crasta OR, Rosenow DT, Mullet JE, Nguyen HT (2000) Molecular mapping of QTLs conferring stay-green in grain sorghum (*Sorghum bicolor* L. Moench). Genome 43:461–469
Zeng ZB (1993) Theoretical basis of precision mapping of quantitative trait loci. PNAS (USA) 90:10972–10976
Zeng ZB (1994) Precision mapping of quantitative trait loci. Genetics 136:1457–1468